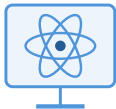


Multi-fidelity problem

self-driving laboratory

candidate pool $\mathcal{X}_{\text{cand}}$

13 benchmarks: 4 synthetic · 9 chemistry & materials



Low fidelity

simulation

cheap, abundant · $\rho \ll 1$



High fidelity

experiment

costly, scarce · cost 1

Controlled comparison

fix budget · schedule · capacity

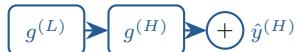
→ **vary only the surrogate**

**Gaussian-process
(GP) family** × 3



MFGP · DKL
sparse variational MFGP

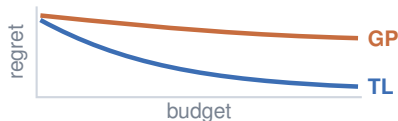
**Transfer-learning
(TL) surrogates** × 9



residual transfer + Bayesian linear-regression head

Transfer-learning wins

on chemistry & materials benchmarks



TL $\geq$ GP on all 9 chemistry & materials pools

compute per optimization loop

GP

TL **up to 45× fewer FLOPs**

closed-loop optimization: propose next query, evaluate, update